Wake2Wake: Feature-Guided Self-Supervised Wave Suppression Method for SAR Ship Wake Detection

Chengji Xu, *Member, IEEE*, Qingsong Wang, Xiaoqing Wang, Xiaopeng Chao, and Bo Pan ,

***Abstract*— Sea clutter and inherent speckle noise in synthetic aperture radar (SAR) images can pose challenges to accurate sea surface target detection, especially for phenomena such as ship wakes reliant on efficient feature extraction. Traditional denoising methods require manual tradeoffs between denoising effects and detail retention. Supervised denoising methods based on deep learning demand a substantial number of real noisy-clean image pairs for training, coupled with specific parameter settings and labeled data amounts. In response to these challenges, this article introduces Wake2Wake, a self-supervised denoising method aimed at enhancing the performance of existing deep learning-based ship wake detectors. The method incorporates a novel ship wake awareness (SWA) block designated to address the distinctive features of turbulent and Kelvin wakes. Furthermore, to overcome the source imbalance problem in the dataset, simulated wake data are integrated into the training process. This not only mitigates dataset imbalances but also significantly improves both denoising and detection performance. The experimental results indicate that Wake2Wake improves the accuracy of Rotated RepPoints by 3.6 mAP and S^2A-Net by 2.6 mAP on the OpenSARWake dataset, respectively. The proposed approach achieves varied extents of improvement, showcasing its potential in mitigating sea clutter and enhancing feature extraction, especially in detecting SAR ship wakes.**

***Index Terms*— Image denoising, self-supervised learning, ship wake detection.**

I. Introduction

SYNTHETIC aperture radar (SAR) is highly valuable for observing oceanic phenomena [1], detecting ships [2], [3], [4], [5], [6], [7], and identifying wakes [8], [9], [10], [11] due to its capabilities of all-weather, all-day observation and high-resolution imaging. Owing to the coherence principle underlying the SAR imaging system, the generation of speckle noise is inevitable [12]. Furthermore, ocean incidents such as huge wind waves and surges on the sea surface significantly impact the image quality, affecting the subsequent process of extracting features and the analysis performance of downstream tasks. SAR ship wake detection faces challenges in achieving optimal model accuracy due to complex and unpredictable environmental disruptions, such as background sea clutter and adverse weather conditions such as strong winds and high waves. To improve the accuracy of SAR ship wake detection, it is crucial to explore techniques for removing noise from SAR images or reducing interference from sea waves.

As a derivative research branch of SAR ship detection, the study of wake detection algorithms helps estimate the ship's speed [13], [14], route [15], or hull information [16]. Traditional algorithms for detecting ship wakes in SAR images rely on techniques such as Randon transform [17], constant false alarm rate (CFAR) detection [18], and their variations. Other algorithms utilize sparse regularization [19] or decomposition [20], as well as morphological analysis, to distinguish the wake structure from the waves individually. In [21], dictionary modeling is employed in morphological analysis to separate the wake structure from the waves. Nevertheless, these techniques, primarily reliant on linear structural characteristics, are prone to misinterpretation due to their resemblance to other comparable formations in SAR images, such as small-scale internal waves and fronts.

The availability of several labeled SAR wake datasets has led to advancements in end-to-end convolutional neural network (CNN) techniques. These advancements have resulted in enhanced efficiency and accuracy in wake detection. For instance, Ding et al. [8] and Del Prete et al. [22] collected 261 and 485 SAR images with ship wakes that were carefully labeled. They compared the effectiveness of the respective detection algorithms using these images. However, the algorithms mentioned above need to extract information regarding the characteristics of the wake directly from SAR images, which inherently contain speckle noise and waves. In addition, their aim is to restore the wake area as accurately as possible to establish precise bounding boxes. The task possesses a certain level of complexity without necessitating additional interventions.

In a previous study, Yuan et al. [12] employed a jointly optimized semantic-aware self-supervised denoising method to enhance the performance of the segmentation task. Similarly, Li et al. [23] integrated dehazing and detection tasks into a unified network, introducing a union architecture and a

Manuscript received 28 January 2024; revised 14 May 2024 and 16 June 2024; accepted 21 June 2024. Date of publication 3 July 2024; date of current version 15 July 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62273365, in part by Xiaomi Young Talents Program, and in part by the Project under Grant 2019ZT08X751. *(Corresponding authors: Qingsong Wang; Xiaoqing Wang.)*

Chengji Xu is with the School of Systems Science and Engineering, Sun Yat-sen University, Guangzhou 510006, China (e-mail: xuchj9@mail2.sysu.edu.cn).

Qingsong Wang, Xiaopeng Chao, and Bo Pan are with the School of Electronics and Communication Engineering, Sun Yat-sen University, Shenzhen 518107, China (e-mail: wangqs5@mail.sysu.edu.cn; chaoxp@mail2.sysu.edu.cn; panb6@mail2.sysu.edu.cn).

Xiaoqing Wang is with the School of Electronics and Communication Engineering, Sun Yat-sen University, Shenzhen 518107, China, and also with the Peng Cheng Laboratory, Shenzhen 518055, China (e-mail: wangxq58@mail.sysu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3422803

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:07:08 UTC from IEEE Xplore. Restrictions apply.

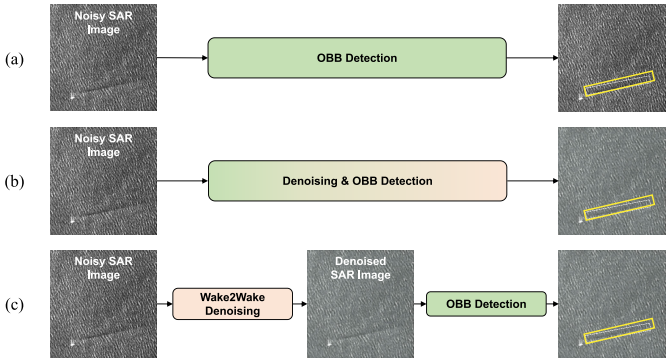


Fig. 1. Three different methods for SAR ship wake detection, where OBB denotes the oriented bounding box, a type of bounding box that is aligned with the object's orientation. (a) Detection model is trained directly on the noisy SAR image, and the results are outputted. (b) SAR image denoising and wake detection tasks are performed simultaneously in a single network, and the detection results with the denoising effect are output directly. (c) Denoising model processes the noisy SAR image and then inputs the detector to train and output the results.

novel robust loss function. Nevertheless, limited research has been conducted to evaluate the effectiveness of enhanced SAR images following restoration processes (such as denoising or wave suppression) on downstream tasks. Building on [23], we hypothesize that the denoising process may impact the detection results of ship wakes.

To address the disparity between high-level SAR ship wake detection and low-level image denoising, we will examine the framework's overall design and the choice of denoising methods. Fig. 1(a) illustrates the process of delivering the unprocessed noisy image to the detector for training and testing. This approach is often used in prior SAR detection algorithms, and there is significant potential for improvement. Fig. 1(b) illustrates a type of jointly optimized framework, represented by Yuan et al. [12] and Li et al. [23]. This approach enhances the performance of image restoration and downstream tasks simultaneously. However, it necessitates a distinct design of the loss function or adding a new module to establish a connection between the two tasks. The third approach, as depicted in Fig. 1(c), employs a two-stage procedure where the detection task and image denoising are performed independently. The benefit of this approach lies in the fact that the refined restoration algorithm can enhance the performance of the detection task to a certain degree. In addition, there are few denoising algorithms available for SAR ship wake. Given different detection algorithms, we will employ a two-stage strategy to design the framework, facilitating subsequent experimental validation. When choosing denoising algorithms, gathering enough clean-noisy image pairs for supervised algorithms that meet the criteria for real SAR wake images is challenging. Therefore, we will explore using self-supervised denoising algorithms that rely on deep learning. These algorithms offer superior generalization capabilities and end-to-end integration compared to conventional methods, such as SAR-BM3D [24], LMMSE-Wavelet [25], and POTDF [26].

To this end, we propose a self-supervised denoising framework named Wake2Wake. This framework can improve the effectiveness of current mainstream-oriented SAR ship wake detectors. To effectively exploit the Kelvin wake and turbulence wake characteristics in SAR images, we have developed a ship wake awareness (SWA) block that concentrates on these two distinct wakes. The module aims to eliminate background waves and noise while enhancing the visibility of the wakes as foreground elements. As a result, we have implemented a more efficient data integration strategy to train Wake2Wake. This strategy involves combining real SAR wake data with simulated data, resulting in improved training efficiency and denoising effectiveness to some extent. These simulated SAR images effectively enhance the singularity and repetitiveness of ship wakes in certain real scenarios. They are well-balanced in terms of radar parameters, sea state parameters, and ship parameters compared to real images. Multiple extensive comparisons and ablation experiments provide evidence that our suggested technology can enhance the performance of current SAR wake detectors to a certain degree. The SAR ship wake dataset that we collected, including rotation detection labels, can be accessed at (https://github.com/libzzluo/OpenSARWake).

II. Proposed Method

This section introduces a feature-guided self-supervised wave suppression module named Wake2Wake. The method takes SAR images containing waves or speckle noise as input aims to produce output images with minimal waves or noise. Wake2Wake can be integrated with various oriented detectors to create a comprehensive framework, resulting in enhanced performance for wake detection using deep learning. This section will provide explicit details.

A. Overall Network Overview

Given that our primary objective is to enhance the performance of current SAR ship wake detectors, we do not allocate effort to create noisy-clean image pairs for supervised denoising learning. Instead, we employ unsupervised denoising methods. Current mainstream unsupervised denoising algorithms primarily utilize pixel masking strategies on input images. This approach requires the network to reconstruct the masked pixels using information from neighboring pixels. Representative examples of such algorithms are NBR2NBR [27] and N2V [28], effectively alleviating the reliance on identity mapping. Nevertheless, these blindspot schemes also suffer from the disadvantage of information loss, as the network cannot perceive the information from the masked pixels. For these reasons, we decided to employ the B2UB [29] network. This network utilizes the original noisy image more efficiently compared to the NBR2NBR and N2V approaches. In addition, it can utilize information from all input pixel points, theoretically preventing any loss of information.

Nevertheless, the level of noise pollution in SAR images surpasses that in optical images. When B2UB is trained exclusively on SAR ship wake images without modification, the denoising results are strikingly inadequate, as demonstrated by our research. The complexity of the sea clutter and noise distribution in SAR ship wake images makes it challenging to perform direct self-supervised denoising of the entire dataset.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:07:08 UTC from IEEE Xplore. Restrictions apply.

However, we drew insights from the advancements in the SAR image task by Yuan et al. [12]. They argue that the application of max pooling in the U-Net architecture of the original self-supervised algorithm is suboptimal, as it tends to generate speckle noise residues in the low-frequency area of the SAR image. This aspect is crucial to consider while addressing ship wake characteristics. Thus, we first modify the training strategy of the self-supervised denoising network and enhance all convolution operations in the original U-Net structure. We also provide an SWA block concentrating on Kelvin wake and turbulent wake characteristics. We refer to this enhanced framework as Wake2Wake.

After the denoised SAR ship wake images are produced by the Wake2Wake network, they are utilized as input for training and testing the ship wake detector. In Section III-B, we analyze the performance variations of Wake2Wake when it is employed on the target detection dataset in the training/validation set, the testing set, and the entire dataset, respectively. In Sections II-B and II-C, we provide the underlying construction principle of the SWA block and a more comprehensive explanation of the precise algorithmic flow of Wake2Wake. To enhance comprehension, we present the pseudo-code for the entire framework flow in Algorithm 1. In Section II-B, we will introduce methods to enhance the original U-Net model, addressing SAR ship wake feature extraction requirements more effectively.

Algorithm 1 Training and Inference Strategy of the Proposed Wake2Wake Denoising Framework and SAR Ship Wake Detector

Step I: Training the Wake2Wake

input : 65 real and 180 simulated SAR ship wake image $\{\mathrm{y}^n, n = 1, 2, \ldots, 245\}$, with different sea states.

output: Well-trained Wake2Wake denoising network $\mathcal{F}_\Omega(\cdot)$

Initialization of trainable parameters for the Wake2Wake network $\mathcal{F}_\Omega(\cdot)$;

for *epoch* $\leftarrow 1$ **to** *max_epoch* **do**

  for $n \leftarrow 1$ **to** 245 **do**

    // Generate masked images patches with blind spots;

    $(\theta_{\mathrm{I}}^{00}, \theta_{\mathrm{I}}^{01}, \theta_{\mathrm{I}}^{10}, \theta_{\mathrm{I}}^{11}) \leftarrow$ `GlobalMasker` (y^n);

    $\theta_y \leftarrow$ `Stack` $(\theta_{\mathrm{I}}^{00}, \theta_{\mathrm{I}}^{01}, \theta_{\mathrm{I}}^{10}, \theta_{\mathrm{I}}^{11})$;

    // Denoiser;

    denoised $\mathrm{y} \leftarrow \mathcal{F}_\Omega(\theta_y)$;

    $\mathcal{M}(\mathcal{F}_\Omega(\mathrm{y})) \leftarrow$ `GlobalMaskMapper` (θ_y);

    // Loss function;

    $L_{rev} = \|\mathcal{M}(\mathcal{F}_\Omega(\mathrm{y})) + \lambda\hat{\mathcal{F}}_\Omega(y) - (\lambda+1)y\|_2^2$

    $L_{reg} = \|\mathcal{M}(\mathcal{F}_\Omega(\mathrm{y})) - y\|_2^2$

    $L = L_{rev} + \eta \cdot L_{reg}$

    Update L *to optimize* $\mathcal{F}_\Omega(\cdot)$;

Inference the full OpenSARWake dataset with $\mathcal{F}_\Omega(\cdot)$*, and the results denoted as* $\hat{\mathrm{x}}$;

Step II: Training the OBB detector

input : Denoised SAR images from OpenSARWake train/val dataset $\{\hat{x}^n, n = 1, 2, \ldots, 3177\}$, with the corresponding oriented bounding box (OBB) labels of ship wakes $\{b_{gt}^n, n = 1, 2, \ldots, 3177\}$

output: Images with detection results

Initialization of trainable parameters for the OBB detector $\mathcal{F}_{det}(\cdot)$ *(i.e. Rotated RepPoints, S*^{2}*A-Net, Oriented R-CNN)*;

for *epoch* $\leftarrow 1$ **to** *max_epoch* **do**

  for $n \leftarrow 1$ **to** 3177 **do**

    $\hat{b}_{gt}^n \leftarrow \mathcal{F}_{det}(\hat{x}^n)$

    $L_{det} \leftarrow$ `CalculateLoss` $(\hat{b}_{gt}^n, b_{gt}^n)$

    Update L_{det} *to optimize* $\mathcal{F}_{det}(\cdot)$;

B. SWA Block

In the broader realm of computer vision, in-context learning [30] has shown notable success in tasks such as image denoising, image enhancement, and image deduplication. Notably, these achievements have been realized without the need for parameter tuning. It seems that refining the structure of CNN does not result in significant advancements in these general domains. However, given the distinct features of the ship wake, a comprehensive solution has yet to emerge in the realm of SAR ship wake recognition. Therefore, we enhance the U-Net architecture utilized in the original B2UB network by considering the unique characteristics of SAR ship wakes. We introduce an SWA block to selectively replace the original convolution operation. Subsequently, we evaluate its Lipschitz constant [31] in comparison to other commonly used attention blocks to assess its generalization capability.

Typically, ship wakes frequently appearing in SAR images consist mainly of turbulent wakes and Kelvin wakes. Our focus is on describing these two types of wakes. Xue et al. [32] employed the frequency channel attention convolution (FcaConv) [33] to investigate the characteristics of Kelvin wakes in optical ship wakes, examining both their image and frequency domains. The experimental results provided evidence that the module effectively captures the optical Kelvin wakes. However, when using the FcaConv with our proposed OpenSARWake [34], we observe that selecting only the discrete cosine transformation (DCT) bases of the top 32 leads to a minor improvement in detection accuracy. Conversely, selecting other frequency components results in a decrease in the original detection accuracy. The absence of Kelvin wakes in SAR images is a common occurrence, with only a turbulent wake, referred to as a black streak, being typically evident. Furthermore, the presence of sea clutter in optical images is usually represented by a Gaussian distribution. However, sea clutter in SAR images is considerably more complicated, posing challenges in differentiating low-frequency ship wakes from the sea clutter. Hence, we opted for an 8×8 DCT basis to rescale the features of the Kelvin wake. In addition, recognizing the imperative need for a mechanism that focuses on the black streak-like turbulent wake, we draw inspiration

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:07:08 UTC from IEEE Xplore. Restrictions apply.

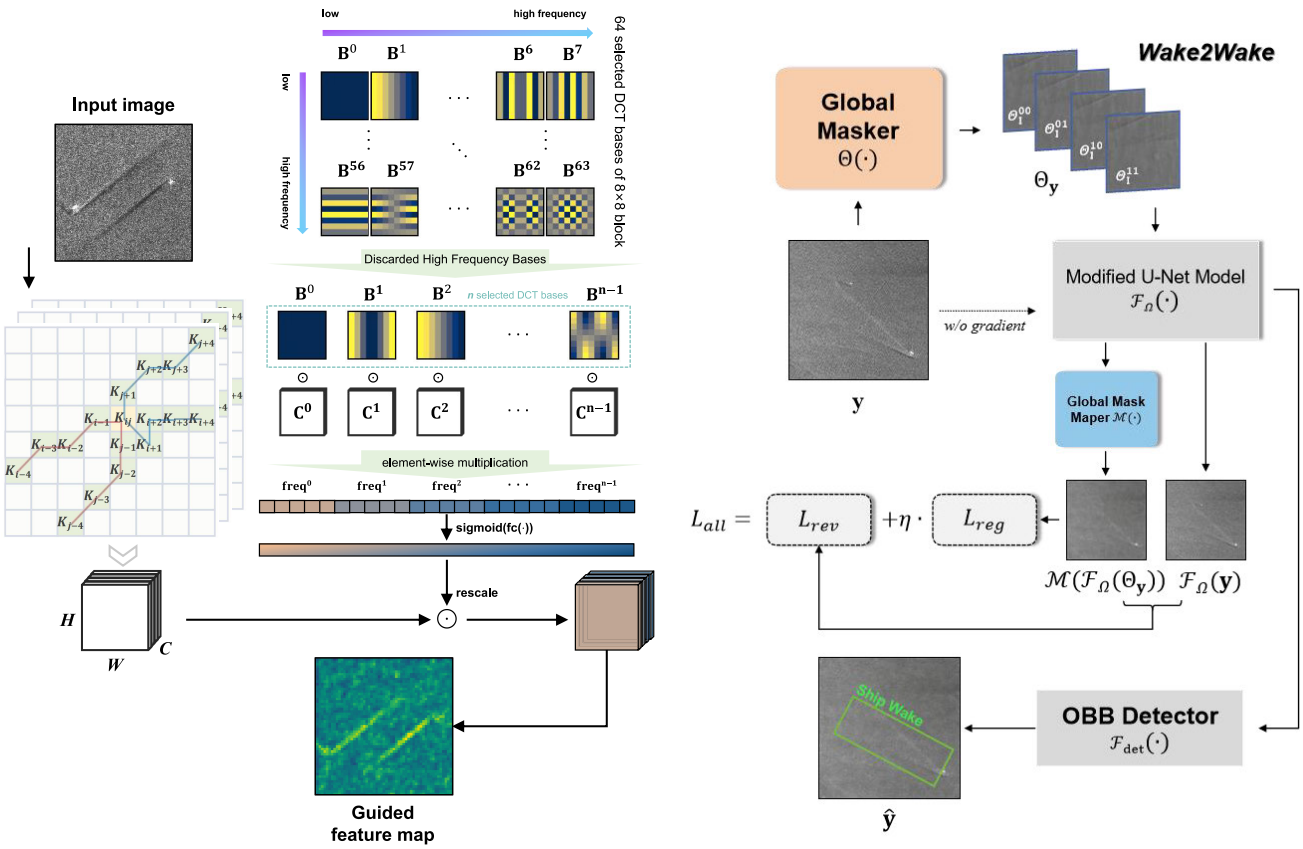


Fig. 2. Illustration of our proposed Wake2Wake framework. (a) Architecture of our proposed SWA block. (b) Feature-guided self-supervised denoising training and inference process. The details regarding the structure mentioned in this figure will be provided in Sections II-B and II-C, correspondingly.

from the successful outcomes achieved by applying elongated tubular convolution structures in the analysis of the digital retinal images for vessel extraction (DRIVE) retina dataset [35] and the Massachusetts Roads dataset [36]. These datasets exemplify the efficacy of such structures in identifying and analyzing linear features across diverse applications, from retinal vessel segmentation to road detection in urban and rural areas. Qi et al. [37] propose employing the dynamic snake convolution (DSConv) method to enhance the U-Net structure's ability to specifically identify the structural characteristics of the turbulent wake with a similar elongated streak. The ultimate architecture of our proposed SWA block is depicted in Fig. 2(a). The modified U-Net structure benefits from the combined application of DSConv and the FcaConv, enabling it to effectively capture the characteristics of both the turbulent wake and the Kelvin wake, unlike the conventional conv block. Initially, the input image undergoes a DSConv operation to extract the features of the turbulence wake. Unlike standard convolution, the position of each grid in DSConv is not a regular square. Taking the x-direction as an example, we represent the position of each grid in DSConv as $K_{i\pm c} = (x_{i\pm c}, y_{i\pm c})$, where $c = \{0, 1, 2, 3, 4\}$ denotes the horizontal position distance of each grid from the center grid. The location selection of the following grid from the center grid is a cumulative operation. Therefore, the entire convolutional grid along the x-direction can be represented as

$$K_{i\pm c} = \begin{cases} (x_{i+c}, y_{i+c}) = \left(x_i + c, y_i + \Sigma_i^{i+c} \Delta y\right) \\ (x_{i-c}, y_{i-c}) = \left(x_i - c, y_i + \Sigma_{i-c}^{i} \Delta y\right). \end{cases} \tag{1}$$

The convolution grid along the y-direction can be represented as

$$K_{j\pm c} = \begin{cases} \left(x_{j+c}, y_{j+c}\right) = \left(x_j + \Sigma_j^{j+c} \Delta x, y_j + c\right) \\ \left(x_{j-c}, y_{j-c}\right) = \left(x_j + \Sigma_{j-c}^{j} \Delta x, y_j - c\right). \end{cases} \tag{2}$$

Typically, Δx and Δy both have decimals, necessitating bilinear interpolation operations

$$K = \Sigma_{K'} I\left(K', K\right) \cdot K' \tag{3}$$

where K denotes the fractional location in (1) and (2), I is the bilinear interpolation kernel, and K' enumerates all the integer spatial locations. I is separated into two 1-D kernels, namely,

$$I\left(K, K'\right) = I\left(K_x, K'_x\right) \cdot I\left(K_y, K'_y\right). \tag{4}$$

Subsequently, the feature map after DSConv, denoted as $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$, is spectrally filtered by DCT. It is categorized into n groups along the channel dimension, with the number of channels in each group as (c/n) (where C is divisible by

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:07:08 UTC from IEEE Xplore. Restrictions apply.

n). The output of the DCT for the different frequencies is

$$\text{freq}^k = \text{DCT}(\mathbf{X}^k) = \sum_{i=0}^{H-1}\sum_{j=0}^{W-1} B_{i,j}^k \cdot \mathbf{X}_{i,j,:}^k \tag{5}$$

where H and W represent the size of the feature map. X_k and $X_{i,j,:}^k$ represent the kth group input feature maps and the X_k element at position (i and j), respectively. $B_{i,j}^k$ is the selected DCT base for the kth group element. After obtaining the 2-D DCT outputs, the feature map $\mathbf{X}_n$ can be rescaled

$$\text{weight} = \text{sigmoid}\left(\text{fc}\left(\text{concat} \times \left(\left[\text{freq}^0, \text{freq}^1, \ldots, \text{freq}^{n-1}\right]\right)\right)\right). \tag{6}$$

Finally, $\mathbf{X}_n$ is multiplied with weight through the Hadamard product to get the final guided feature map

$$\tilde{\mathbf{X}} = \text{weight} \odot \mathbf{X}. \tag{7}$$

In our practical experiments, we replaced all the plain convolutions in the original U-Net with SWA_BLOCK, and the final improved U-Net structure is shown in Table I.

To quantitatively analyze the robustness of our suggested SWA_BLOCK, we computed its Lipschitz constant and compared it with other common attention blocks. While accurately determining the Lipschitz constant of a multilayer network is a nondeterministic polynomial-time hard (NP-hard) problem, we can make an approximation by calculating an upper bound on the Lipschitz constant of the network. For instance, a convolutional block in a typical ResNet can be represented as

$$f(x) = W_1\phi(W_0 x + b_0) + b_1 \tag{8}$$

where x is an input to this block. The weight matrix and bias vector of the two neighboring convolutional layers are W_0, b_0 and W_1, b_1, respectively. ϕ represents the concatenation of activation functions, where $\phi(x) = [\varphi(x_1), \ldots, \varphi(x_n)]^\top$.

For all x, $y \in \mathbb{R}$, setting the Lipschitz constant of the block to be $\sqrt{L}$, its Lipschitz condition is expressed as

$$\|f(x) - f(y)\|_2 \le \sqrt{L}\|x - y\|_2. \tag{9}$$

Squaring both sides of the inequality in (9) gives

$$(f(x) - f(y))^\top (f(x) - f(y)) \le L(x - y)^\top (x - y). \tag{10}$$

According to the slope-restricted nonlinearity definition in [38], the function φ is slope-restricted on $[\alpha, \beta]$, where $0 \le \alpha < \beta < \infty$ if

$$\alpha \le \frac{\varphi(y) - \varphi(x)}{y - x} \le \beta \quad \forall x, y \in \mathbb{R}. \tag{11}$$

To understand the connection between incremental quadratic constraints and slope-restricted nonlinearity, (11) can equivalently be expressed as the following single inequality:

$$\left(\frac{\varphi(y) - \varphi(x)}{y - x} - \alpha\right)\left(\frac{\varphi(y) - \varphi(x)}{y - x} - \beta\right) \le 0. \tag{12}$$

TABLE I

WELL-DESIGNED U-NET NETWORK ARCHITECTURE IN OUR EXPERIMENTS. IN ORDER TO ADDRESS THE SPECKLE NOISE RESIDUALS IN THE LOW-FREQUENCY AREA OF THE SAR IMAGE DURING DENOISING, WE SUBSTITUTED THE ORIGINAL MAXPOOL WITH AVGPOOL. THE STRUCTURE OF THE SWA_BLOCK IS FURTHER EXPLAINED IN SECTION II-B

Layer	F_{out}	Function
INPUT	1	Noisy SAR Image
SWA_BLOCK0	48	
SWA_BLOCK1	48	
POOL1	48	Avgpool 2×2
SWA_BLOCK2	48	
POOL2	48	Avgpool 2×2
SWA_BLOCK3	48	
POOL3	48	Avgpool 2×2
SWA_BLOCK4	48	
POOL4	48	Avgpool 2×2
SWA_BLOCK5	48	
POOL5	48	Avgpool 2×2
SWA_BLOCK6	48	
UPSAMPLE5	48	Upsample 2×2
CONCAT5	96	Concat output of POOL4
SWA_BLOCK7	96	
SWA_BLOCK8	96	
UPSAMPLE4	96	Upsample 2×2
CONCAT4	144	Concat output of POOL3
SWA_BLOCK9	96	
SWA_BLOCK10	96	
UPSAMPLE3	96	Upsample 2×2
CONCAT3	144	Concat output of POOL2
SWA_BLOCK11	96	
SWA_BLOCK12	96	
UPSAMPLE2	96	Upsample 2×2
CONCAT2	144	Concat output of POOL1
SWA_BLOCK13	96	
SWA_BLOCK14	96	
UPSAMPLE1	96	Upsample 2×2
CONCAT1	99	Concat INPUT
SWA_BLOCK15	96	
SWA_BLOCK16	96	
OUTPUT	1	Denoised SAR Image

By multiplying by $(y - x)^2$ and rearranging, we can express (12) in the following form:

$$\begin{bmatrix} x - y \\ \varphi(x) - \varphi(y) \end{bmatrix}^\top \begin{bmatrix} -2\alpha\beta & \alpha + \beta \\ \alpha + \beta & -2 \end{bmatrix} \begin{bmatrix} x - y \\ \varphi(x) - \varphi(y) \end{bmatrix} \ge 0. \tag{13}$$

The definition of $T \in \mathcal{T}_n$ can be found in [38], supposing there exists $L > 0$ such that the following symmetric matrix inequality holds:

$$M(L, T) := \begin{bmatrix} -2\alpha\beta W_0^\top W_0 - LI_n & (\alpha + \beta)W_0^\top T \\ (\alpha + \beta)T W_0 & -2T + W_1^\top W_1 \end{bmatrix} \preceq 0 \tag{14}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:07:08 UTC from IEEE Xplore. Restrictions apply.

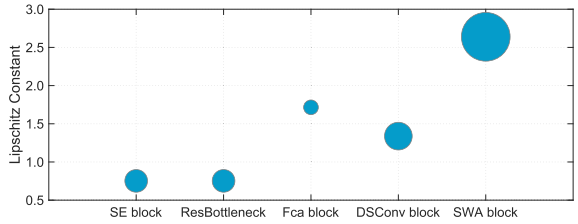


Fig. 3. Comparison of Lipschitiz constants in different attention modules. The size of the radius of the circular symbols is proportional to the computed FLOPS of the module. It is important to note that a larger Lipschitiz constant does not necessarily imply more robustness; on the contrary, a larger Lipschitiz constant will make the training of the module more unstable. Therefore, Lipschitiz constants are used here solely for theoretical analysis of their role in the module.

where I_n is the n-dimensional identity matrix. At this point, (13) is always satisfied. Hence, when $M(L,T) \preceq 0$, the ResBottleneck will satisfy the Lipschitz condition, taking $\sqrt{L}$ as the Lipschitz constant. This NP-hard problem can be transformed into the following semidefinite programming problem (SDP):

$$\min L, \quad \text{s.t. } M(L,T) \preceq 0 \text{ and } T \in \mathcal{T}_n. \tag{15}$$

Eventually, we employed the LipSDP [38] algorithm to approximate the Lipschitz constants of various common attention blocks and determine the computational floating point operations per second (FLOPS), as depicted in Fig. 3.

C. Self-Supervised Denoising Framework Wake2Wake

We selected B2UB as the baseline method for several reasons. First, it is a self-supervised denoising technique that only requires a noisy SAR image. Second, it effectively utilizes all the information in the original noisy image. Finally, it applies a more stable Re-visible loss function throughout the training phase. However, most self-supervised denoising methods were initially developed for optical images [39], [40]. These methods do not consider the distinctive features of SAR sea images, particularly ship wakes such as Kelvin and dark streak-like turbulent wakes. Waves and sea clusters often obscure these wakes. In Sections II-A and II-B, we outline the enhancements that we have implemented to achieve this objective.

Our primary objective is to integrate the trained Wake2Wake model into the OBB detector. Typically, we use the entire detection dataset as inputs for the denoising network. However, challenges arise in simultaneously learning all noise distributions in the detection dataset without increasing the network's capacity, often necessitating refined parameterization. A comprehensive explanation of the collection and preprocessing of the OpenSARWake dataset is provided in Section III-A. To address this, we propose utilizing a smaller yet more feature-enriched set of training samples. This subset comprises 65 carefully selected SAR images featuring prominent wind waves and wakes from the original OpenSARWake dataset. In addition, we include 180 simulated SAR images generated using a total of 60 sea conditions with the full-link ocean surface SAR imaging simulator [41], [42]. This simulation covers five distinct wind speeds ranging from 5.5 to 13.5 m/s and 20 wind directions at intervals of 30°. To ensure a thorough evaluation, three different bands of SAR parameters were employed, resulting in diverse and extensive characterizations. For further details on the acquisition and generation principles of this specific data subset, please refer to Section III-A.

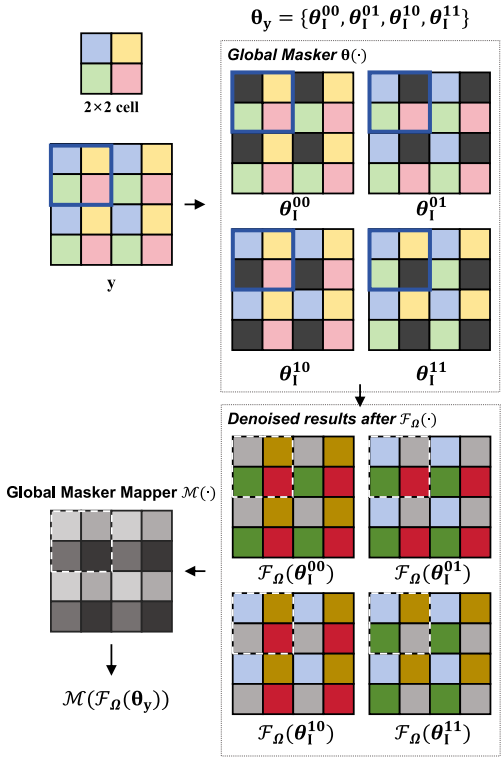


Fig. 4. Implementation detail of the Global Mask $\theta(\cdot)$ and Global Mask Mapper $\mathcal{M}(\cdot)$. In the experiments of this article, 2 × 2 mask cells are used throughout.

Hence, the input of our newly designed Wake2Wake network is represented by $\{y^n, n = 1, 2, \ldots, 245\}$, and the dimensions of the images are uniformly set to 1024 × 1024 to maintain a wide range of intricate features. As depicted in Figs. 2(b) and 4, each y^n is initially transformed via the Global Masker $\Theta(\cdot)$ to produce four masked image patches $(\theta_I^{00}, \theta_I^{01}, \theta_I^{10}, \theta_I^{11})$. The procedure is given as follows.

1) The input image y^n is partitioned into $\lfloor W/s \rfloor \times \lfloor H/s \rfloor$ separate cells, each with a size of $s \times s$. For our experiment, we fixed the W and H values to 1024. When is greater than 2, the computational load of the masking operation increases geometrically; therefore, s is set to 2 in this article, as presented in Fig. 4.
2) Before being submitted to the Global Mask Mapper $\mathcal{M}(\cdot)$ for sampling, $(\theta_I^{00}, \theta_I^{01}, \theta_I^{10}, \theta_I^{11})$ must be organized into a masked volume θ_y and processed by our modified U-Net model $\mathcal{F}_\Omega(\cdot)$ and Global Mask Mapper $\mathcal{M}(\cdot)$ to obtain $\mathcal{M}(\mathcal{F}_\Omega(y))$.
3) In another sequence, the initial noisy image y^n is immediately fed into the $\mathcal{F}_\Omega(\cdot)$ to obtain the denoised output $\mathcal{F}_\Omega(y)$.

The loss function is defined as follows during the training process of Wake2Wake:

$$L_{\text{rev}} = \left\| \mathcal{M}(\mathcal{F}_\Omega(y)) + \lambda \hat{\mathcal{F}}_\Omega(y) - (\lambda+1)y \right\|_2^2$$
$$L_{\text{reg}} = \left\| \mathcal{M}(\mathcal{F}_\Omega(y)) - y \right\|_2^2$$
$$L = L_{\text{rev}} + \eta \cdot L_{\text{reg}}.$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:07:08 UTC from IEEE Xplore. Restrictions apply.

The hyperparameter η is a constant governing the initial impact of the blind term and the stability of the training process. In contrast, the hyperparameter λ is a variable regulating the strength of visible components during the conversion from blind to unblind. The starting and ending values of λ are denoted as λ_1 and λ_2, respectively. We determined the values of α, λ_1, and λ_2 through empirical observation, setting η to 1, λ_1 to 2, and λ_2 to 20.

Following the completion of Wake2Wake training, the denoised OpenSARWake dataset is employed to train various types of OBB detectors. For further details on this process, please refer to the pseudocode in Algorithm 1.

III. Experimental Results and Discussion

A. Details of Datasets and Implementation

To validate the effectiveness of our proposed Wake2Wake method, it is imperative to train and evaluate it using an extensive dataset of SAR ship wakes. However, there is a scarcity of publicly available benchmarks in this domain. Table II demonstrates that Del Prete et al. [22] and Ding et al. [8] have gathered two SAR ship wake datasets of a certain scale in the C-band. However, their datasets are limited since they only cover the C-band and are inaccessible to the public. Consequently, the available SAR ship wake images are insufficient to meet the requirements of data-driven deep learning networks. In order to facilitate Wake2Wake training and subsequent deep learning-based SAR ship wake detection, we constructed an OpenSARWake dataset. Table II demonstrates that we considered the diversity of ship wake characteristics in various frequency bands, including X-, C-, and L-bands. To accomplish this, we gathered SAR images with ship wakes from these bands from offshore regions, as depicted in Fig. 5(a). This facilitated the collection of the related automatic identification system (AIS) data, which can be utilized for potential subsequent investigations. Fig. 5(b)–(e) illustrates the statistical distribution of our proposed OpenSARWake dataset. These images were obtained from https://search.asf.alaska.edu/ and https://earth.esa.int/eogateway/catalog/.

We collected ground-range detected (GRD) products for the C-Band Sentinel and H2.2-level products for the L-Band ALOS PALSAR images. For X-Band TerraSAR-X images, we primarily used single-look complex (SLC) products. To maintain consistent quality, we utilized the SNAP software to process the data and generate geocoded ellipsoid-corrected (GEC) products. In addition, essential preprocessing techniques, such as the adaptive contrast algorithm, were applied, and the image was ultimately resized to dimensions of 1024 × 1024. Expert labeling ensured meticulous annotations of all rotated bounding boxes for reliability. To validate the annotations for the ship wakes acquired by ALOS-PALSAR and Sentinel-1A near Denmark, we used AIS data obtained from https://web.ais.dk/aisdata/, and the visual inspection was used for the remaining images. The OpenSARWake dataset consists of 653 C-band images, 299 X-band images, and 2221 L-band images. In addition, there were 231 images containing Kelvin wake and 2841 images containing turbulent wake. A random selection process allocated 60% of the total images to the training set, while 20% were assigned to both the validation and test sets. During the training of the Wake2Wake component, it was observed that using the complete OpenSARWake training set resulted in significantly unsatisfactory denoising effects, rendering it almost ineffective on the original SAR images. The presence of sea wave interference and other oceanic phenomena, along with speckle noise, posed challenging in denoising and wave depression, surpassing those encountered in optical benchmarks such as BSD300 [43] and Kodak [44]. Therefore, establishing a more efficient and representative feature space became necessary to enhance the denoising effect of Wake2Wake.

TABLE II
Published Datasets in SAR Ship Wake Detection Community. HBB: Horizontal Bounding Box; OBB: Oriented Bounding Box; and PMA: Polygon Mask Annotation

	SSWD [22]	Gaofen-3 dataset [8]	OpenSARWake [34]
Images	261	485	3973
Image Size	512×512	≈800	1,024×1024
Source	Sentinel-1	Gaofen-3	TerraSAR-X Sentinel-1 ALOS-PALSAR
Imaging modes	IW	UFS, FS-I, FS-II	SM (X-Band) IW, SM (C-Band) FBD (L-Band)
Category	1	2	1
Polarization	VV	Single, Dual	HH, VV
Annotation	PMA	HBB	OBB

Not all sea surface SAR images exhibit significant wave activity in real life. However, these images typically show distinct turbulence wakes with greater clarity. Thus, it is expected that Wake2Wake can emphasize denoising and effectively suppress waves to enhance the detection performance of SAR ship wakes that are affected by significant wave interference. While constructing the feature space, namely, the data distribution, for training Wake2Wake, we deliberately selected 65 SAR images with clearly visible waves and ship wakes from the OpenSARWake dataset. The OpenSARWake collection comprises 408 scenes of SAR images from ALOS PALSAR, 120 scenes of images from Sentinel 1, and only 55 scenes of images from TerraSAR-X. It is important to note that ship wakes in different bands have their own unique characteristics. Therefore, if there are too few or too many images in a particular band, it may cause the denoiser to learn biased wake features. However, this issue can be mitigated to some extent by adding simulated SAR wake images. In addition, we integrated 180 simulated SAR ship wake images. Commonly observed ship wake types in SAR images include Kelvin wake, turbulence wake, internal wave wake, and narrow-V wake [45]. Owing to their infrequency, we have excluded the internal wave wake and narrow-V wake from our simulation. The Kelvin wake refers to a modeling method based on the potential flow theory proposed by [46]. Based on this method, we also consider the sea surface motion, the time-varying characteristics of the scattering units, and the decoherence effect of the sea surface in the echo simulation. The simulator used for this task can be found at https://github.com/SYSUSARSimu/KWFull-Link_SARSim. To model the turbulent wake, we employ our simulation strategy [41] based on the energy spectrum balance equation. This method has the additional advantage of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:07:08 UTC from IEEE Xplore. Restrictions apply.

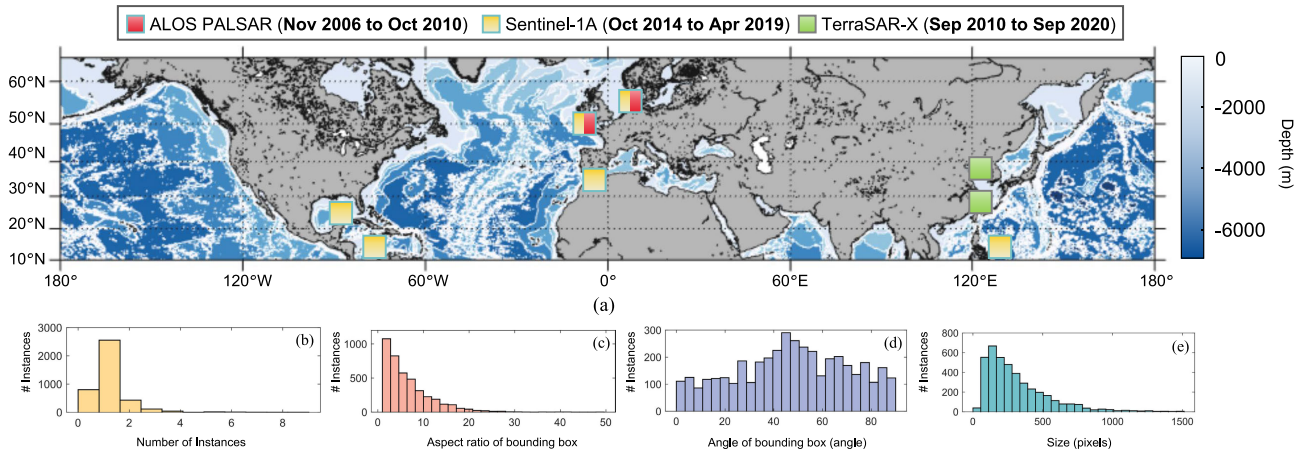


Fig. 5. (a) Geographic location and temporal distribution of the collected OpenSARWAke dataset applied in our study. (b)–(e) Statistical distribution of the instances in the OpenSARWake dataset.

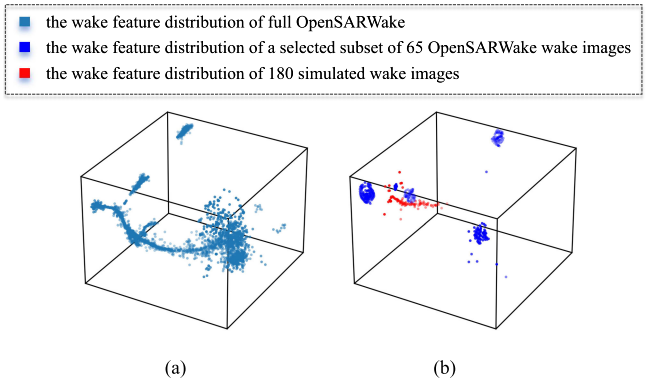


Fig. 6. Effects of the datasets' samples on manifold distributions. (a) OpenSARWake dataset. (b) Proposed multisource dataset.

accurately simulating the turbulent wake across a distance that could exceed kilometers. We simulate SAR images in three frequency bands: X-band, C-band, and L-band. The choice of SAR parameters is guided by that used in [47]. To enhance the variety of the simulated ship wakes, we utilize five different wind speeds ranging from 5.5 to 13.5 m/s and 12 wind directions with intervals of 30° for the sea surface simulation. The simulation approach employed for the sea surface is based on a full-link ocean surface SAR imaging simulator [42]. Fig. 4 demonstrates the application of the unified manifold approximation and projection (UMAP) method to analyze the OpenSARWake dataset and our proposed mixed datasets. All the SAR images were projected into a 3D space, revealing a wide distribution in the manifold space. In Fig. 6(a), the cluster of feature points on the right-hand side indicates that most wake images share similar characteristics. In Fig. 6(b), the red and blue points represent the features of a selected subset of 65 OpenSARWake wake images and 180 simulated wake images, respectively. Although the total number of these images is smaller than the complete OpenSARWake dataset, they still occupy a similar space in the manifold distribution. This highlights the richness of the characteristics in the simulated wake images. Thus, to summarize, we employ a small-scale mixed-source SAR ship wake dataset to train Wake2Wake. However, the training and testing datasets utilized in our comprehensive framework comprise the whole OpenSARWake dataset.

B. Comparison With State-of-the-Art Methods

To evaluate the performance disparity between our proposed Wake2Wake method and other denoising algorithms on various detectors, we specifically selected two categories of SAR image denoising algorithms for comparison. These include traditional methods such as LMMSE-Wavelet [25], SAR-BM3D [24], FANS [48], POTDF [26], and PPB [49], as well as deep learning-based methods such as SAR-CNN [50], NBR2NBR [27], and B2UB [29]. For the rotated detectors, we selected anchor-free methods such as Rotated RepPoints [51], single-stage methods such as S^2A-Net [52], and two-stage methods such as Oriented R-CNN [53] for cross-comparison. We selected an equivalent number of looks (ENL), peak signal-to-noise ratio (PSNR), and structural similarity index (SSIM) for their recognized efficacy in similar SAR despeckling tasks, evaluating homogeneity, reconstruction quality, and structural integrity, respectively. These metrics are complemented by recall and mean average precision (mAP) to assess the overall impact on detection performance. While other metrics such as TCR [55] and EPD-ROA [56] were considered, we prioritized those with a direct, established correlation to despeckling effectiveness and downstream detection capabilities. The experiments were conducted using NVIDIA A6000 GPUs. For the denoising algorithms other than Wake2Wake, we used the same hyperparameters as specified in their respective original papers. For the training process of the Wake2Wake algorithm, the initial learning rate was set to $3e^{-4}$, with a weight decay of $1e^{-8}$. We trained the model for 200 epochs with a batch size of 2, using the Adam optimizer for learning rate updates. Regarding the detection algorithms Rotated RepPoints [51], S^2A-Net [52], and Oriented R-CNN [53], their training hyperparameters can be referred to in [34]. Notably, despite our Wake2Wake approach, all other methods were trained and evaluated using the complete OpenSARWake dataset.

The comparative analysis of several denoising algorithms on OpenSARWake is presented in Table III. The ENL is frequently employed to quantify the speckle suppression capabilities of various SAR image filters. A higher ENL value indicates effective smoothing. Traditional methods, POTDF and PPB, exhibit significantly higher ENL values than deep

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:07:08 UTC from IEEE Xplore. Restrictions apply.

TABLE III

DENOISING EFFECTS AND QUANTITATIVE COMPARISON OF THE DIFFERENT DENOISING METHODS ON THE OPENSARWAKE DATASET. THE PERFORMANCE FOR THE TRAINING, VALIDATION, AND TEST SETS IS SHOWN, RESPECTIVELY. BOLD AND UNDERLINE INDICATE THE BEST RESULTS

Metric	Data	LMMSE-Wavelet [25]	SAR-BM3D [24]	FANS [48]	POTDF [26]	PPB [49]	SAR-CNN [50]	NBR2NBR [27]	B2UB [29]	Wake2Wake
ENL	# Train	76.57	51.86	92.32	147.85	141.98	19.42	24.42	22.37	24.99
	# Val	81.51	53.16	100.91	168.13	161.24	19.62	40.31	24.43	25.81
	# Test	73.19	48.72	90.23	149.18	136.60	18.91	39.03	21.13	24.07
	Avg.	77.09	51.25	94.49	**155.05**	146.61	19.32	**34.59**	22.64	24.96
PSNR	# Train	60.94	62.59	77.09	75.82	79.80	31.08	29.85	30.48	31.26
	# Val	61.96	64.69	76.79	74.79	78.07	31.11	27.59	29.26	31.04
	# Test	59.65	62.40	76.68	74.56	77.93	31.03	27.77	31.16	31.53
	Avg.	60.85	63.23	76.85	75.06	**78.60**	31.07	28.40	30.30	**31.28**
SSIM	# Train	0.582	0.610	0.650	0.614	0.635	0.743	0.818	0.849	0.894
	# Val	0.596	0.623	0.662	0.603	0.629	0.745	0.699	0.808	0.846
	# Test	0.571	0.601	0.657	0.593	0.622	0.739	0.699	0.853	0.842
	Avg.	0.583	0.611	**0.656**	0.603	0.629	0.742	0.739	0.837	**0.861**

TABLE IV

COMPARISONS OF DETECTION PERFORMANCE ON THE OPENSARWAKE TEST SET. TYPE I INDICATES DENOISING PROCESSING APPLIED ONLY TO THE TRAIN/VALIDATION SET, WHILE TYPE II SIGNIFIES APPLICATION SOLELY TO THE TEST SET. TYPE III DENOTES DENOISING OF THE ENTIRE DATASET. THE RESULTS IN THIS TABLE REPRESENT DETECTION **RECALL** AND **MAP**, RESPECTIVELY. BOLD AND UNDERLINE INDICATE THE BEST RESULTS

Detector	Rotated RepPoints [51]			S^2A-Net [52]			Oriented R-CNN [53]		
Type	I	II	III	I	II	III	I	II	III
Noisy	51.3/37.8			46.9/33.8			**53.8/42.5**		
LMMSE-Wavelet [25]	35.1/19.3	44.8/25.4	57.3/38.0	20.5/15.8	33.1/23.7	46.9/33.2	22.7/16.6	38.3/25.8	43.6/33.2
SAR-BM3D [24]	42.7/19.7	47.4/25.4	58.6/36.3	31.0/21.0	34.6/25.0	45.3/32.3	28.8/18.4	39.1/26.0	44.7/33.5
FANS [48]	28.6/15.5	32.8/17.7	54.1/35.1	19.4/13.3	22.8/17.1	42.3/32.5	18.9/12.1	22.8/17.2	37.7/27.0
POTDF [26]	34.1/14.2	30.8/19.3	52.7/33.2	22.5/16.1	18.6/13.4	44.4/32.4	20.8/16.6	20.6/17.4	39.8/26.9
PPB [49]	19.4/11.5	21.3/12.5	51.4/33.1	11.6/11.5	11.0/11.5	41.7/31.5	8.4/6.1	13.0/12.1	40.8/31.8
SAR-CNN [50]	27.1/13.4	7.9/2.7	36.7/18.2	20.6/14.8	4.2/4.1	24.2/16.5	21.5/15.3	4.0/9.1	21.8/15.0
NBR2NBR [27]	51.1/30.3	45.1/23.9	48.8/28.9	42.1/30.4	29.0/19.4	34.7/24.5	36.0/25.6	35.6/25.7	32.6/25.2
B2UB [29]	51.6/29.9	53.8/30.2	58.1/34.0	43.4/29.9	42.1/31.9	45.1/31.7	39.2/26.0	48.8/35.7	43.1/32.2
Wake2Wake	54.2/37.5	56.5/39.0	**59.1/41.4**	44.0/32.3	47.3/35.2	**48.9/36.4**	50.7/38.7	51.7/40.2	50.2/37.9

learning methods. However, we argue that this does not necessarily enhance the accuracy of ship wake detection. Table IV shows that the higher ENL values of POTDF and PPB decrease their mAP on detection algorithms, especially for Rotated RepPoints, which is an anchor-free object detector. Nevertheless, both methods enhance their detection recall when employing Type III training strategies. Furthermore, our examination of the ENL values of the deep learning algorithms did not reveal significant regularities.

PSNR and SSIM quantify the image quality by comparing the maximum signal to background noise and by evaluating image similarity before and after denoising, respectively. Similarly, in conventional methods, higher PSNR and SSIM values do not necessarily indicate higher mAP values. In the case of B2UB and Wake2Wake, slightly higher PSNR and SSIM values result in higher recall and mAP values compared to other deep learning methods. However, these methods are constrained by the limitations of their training strategy, as mentioned in Section III-A, which restricts their potential. Our proposed Wake2Wake method surpasses all other denoising algorithms on the detection accuracy of Rotated RepPoints and S^2A-Net. This is achieved through an enhanced training strategy and the utilization of a customized SWA block designed specifically for ship wake features. Notably, our method significantly outperforms the results obtained without the application of a denoising algorithm. Currently, the consensus is that the application of traditional denoising algorithms tends to decrease the accuracy of all the detection methods listed in Table IV. This is primarily due to the fact that traditional denoising algorithms rely on a fixed approach to image processing, which lacks the capacity to adapt and generalize well. In addition, these algorithms are incapable of retaining and acquiring information regarding the characteristics present in the data. Furthermore, if not enhanced, the denoising techniques that rely on deep learning also diminish their effectiveness on the aforementioned three categories of detection algorithms. Overfitting and incomplete feature learning can negatively impact the model's performance on the detector. In addition, disregarding ship wake features may reduce the effectiveness of the method. Our proposed Wake2Wake algorithm, built upon the research of previous scholars, addresses these issues.

Furthermore, we provide a visual comparison of the denoising and detection results from a test sample image, as depicted in Figs. 7–9. It is evident that the conventional methods for suppressing waves and noise lack stability, as both the POTDF and PPB algorithms exhibit some degree of oversmoothing in Fig. 7. Conversely, the deep learning algorithms demonstrate

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:07:08 UTC from IEEE Xplore. Restrictions apply.

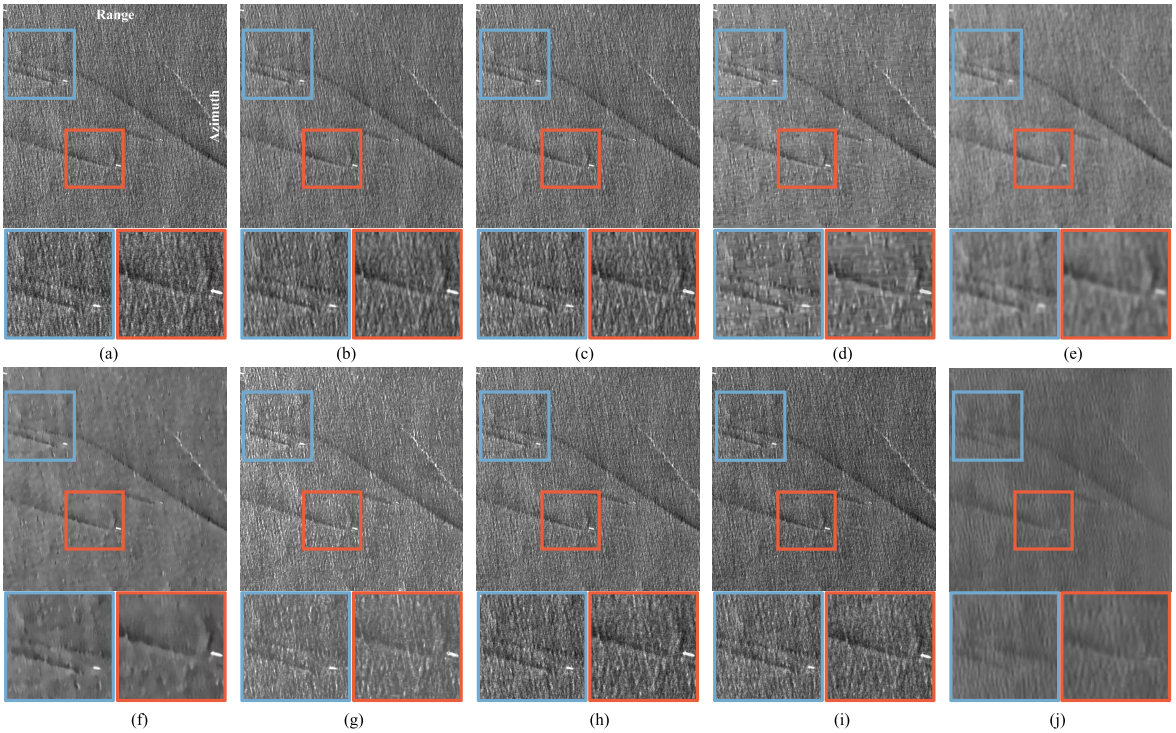


Fig. 7. Visual comparison of SAR image denoising results for the OpenSARWake test sample image using different methods. The image used was acquired through the C-band Sentinel-1A satellite. It is of the English Channel and is a ground-range detected (GRD) product acquired in StripMap mode with a pixel spacing of 10 m. The wind velocity is 6.7 m/s and the wind direction is 74.4°, in which counting goes from the azimuth direction clockwise. The sea state information was derived from the ERA5 hourly data on single levels [54]. (a) Nosiy SAR image. (b) LMMSE-Wavelet. (c) SAR-BM3D. (d) FANS. (e) POTDF. (f) PPB. (g) SAR-CNN. (h) NB2NB. (i) B2UB. (j) Wake2Wake.

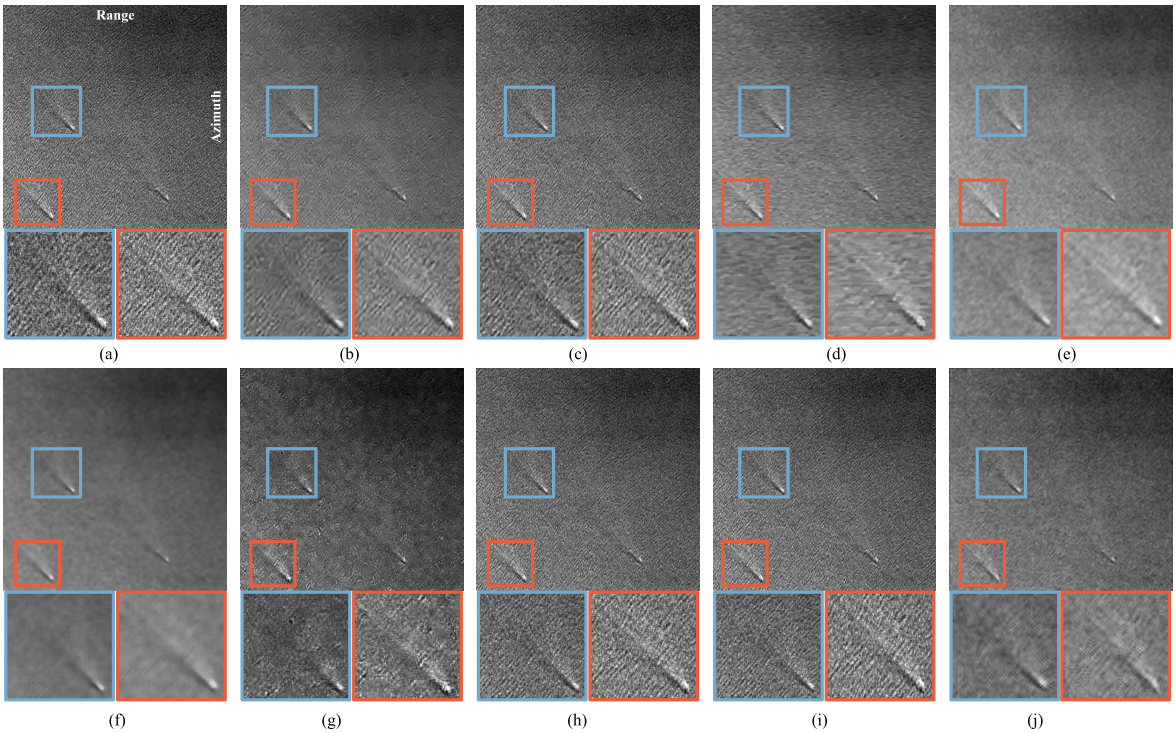


Fig. 8. Visual comparison of SAR image denoising results for the OpenSARWake test sample image using different methods. The image used was acquired through the C-band Sentinel-1A satellite. It is of the English Channel and is a ground-range detected (GRD) product acquired in StripMap mode with a pixel spacing of 10 m. The wind velocity is 8.4 m/s and the wind direction is 255.5°, in which counting goes from the azimuth direction clockwise. The sea state information was derived from the ERA5 hourly data on single levels [54]. (a) Nosiy SAR image. (b) LMMSE-Wavelet. (c) SAR-BM3D. (d) FANS. (e) POTDF. (f) PPB. (g) SAR-CNN. (h) NB2NB. (i) B2UB. (j) Wake2Wake.

a certain level of overfitting. In contrast, our Wake2Wake algorithm effectively suppresses waves while preserving wake features, leading to improved accuracy in bounding-box localization, as demonstrated in Fig. 9.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:07:08 UTC from IEEE Xplore. Restrictions apply.

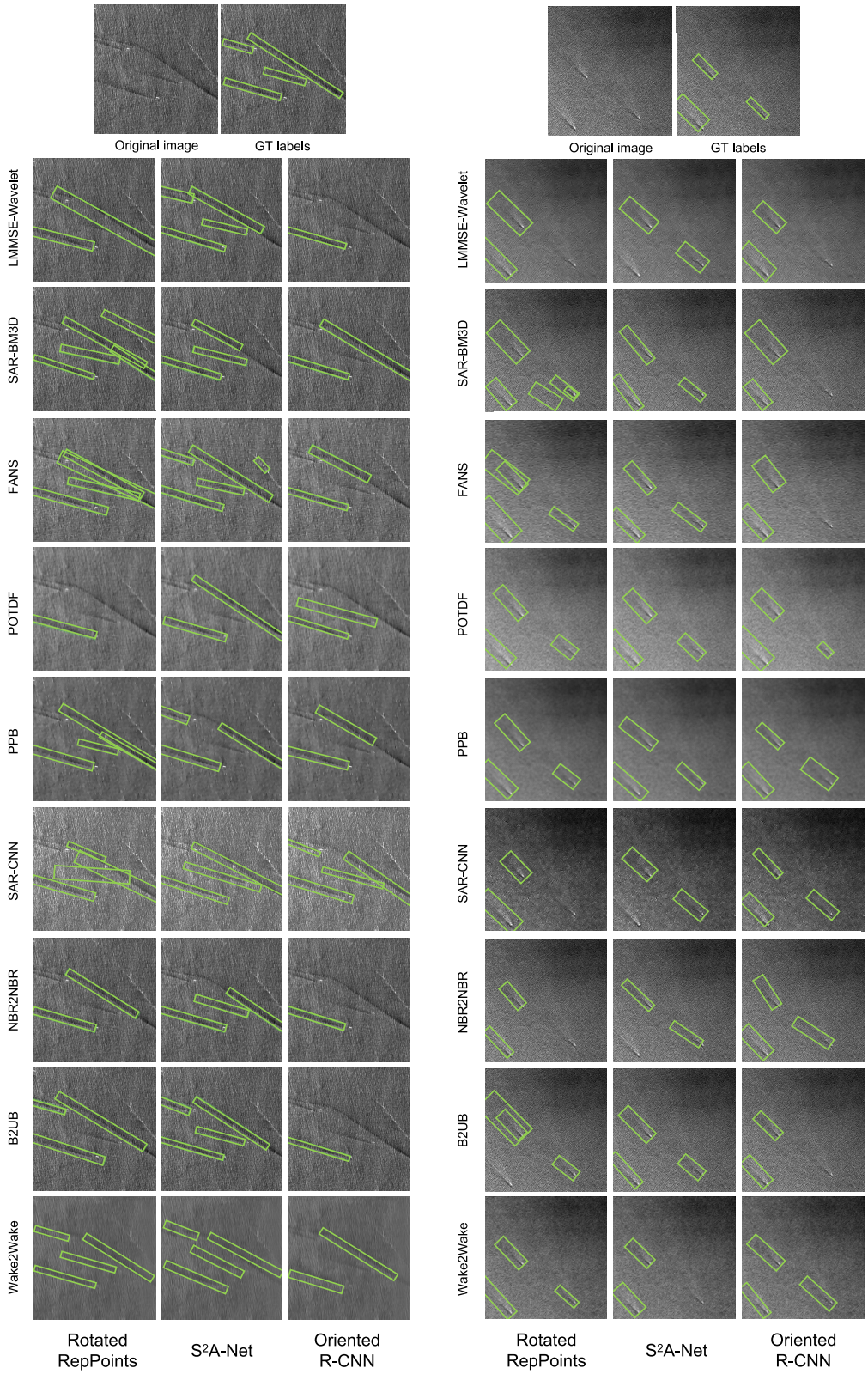


Fig. 9. Detection results on the OpenSARWake test sample image by different denoising and detection methods with (a) corresponding to Fig. 7 and (b) corresponding to Fig. 8. Here, the GT labels represent the ground-truth labels.

C. Ablation Studies

In this section, we conducted ablation experiments to investigate the influence of the dataset used to train Wake2Wake and the convolution operation applied in the SWA block to focus on two distinct ship wake features in the final detection results. The experiments followed the Type III strategy outlined in Table IV, where the training, validation, and test sets for training the OBB detector were all denoised.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:07:08 UTC from IEEE Xplore. Restrictions apply.

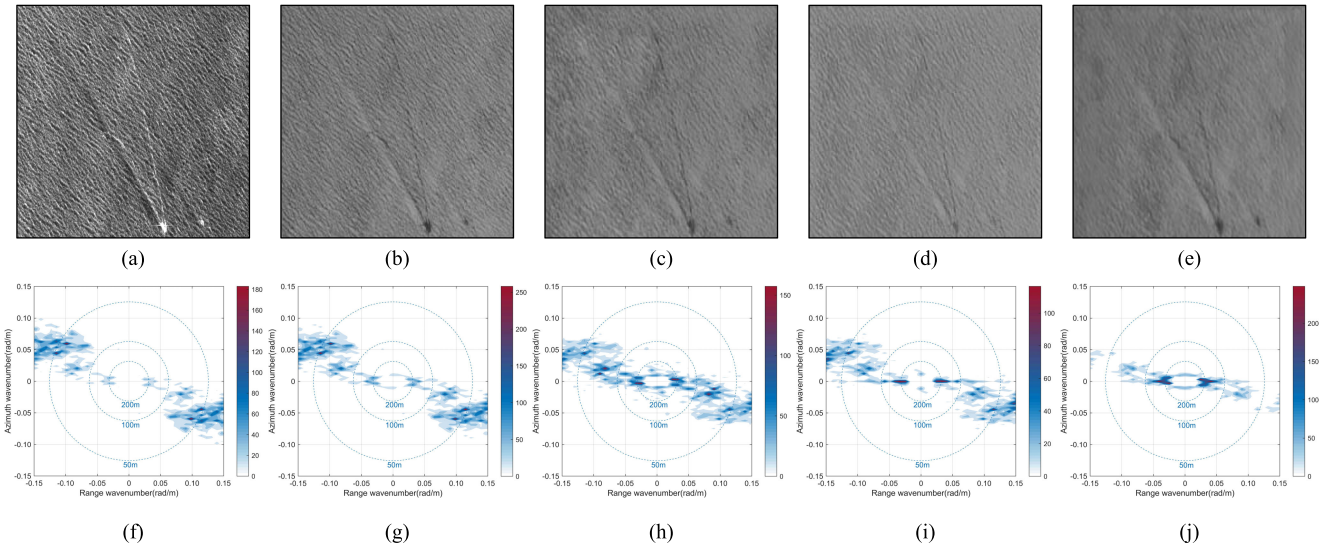


Fig. 10. Comparison of the effect of FcaConv and DSConv, which are utilized in our proposed SWA block. The first row shows the results after denoising with different convolution operations, and the second row shows the corresponding SAR image spectrum. (a) and (f) Noisy SAR image. (b) and (g) Effect of plain convolution. (c) and (h) Effect of FcaConv. (d) and (i) Effect of DSConv. (e) and (j) Effect of our proposed SWA block.

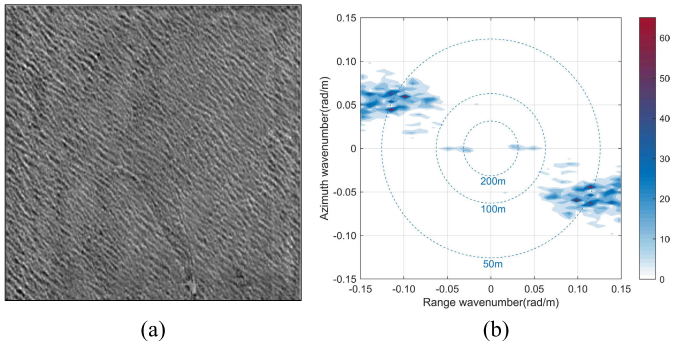


Fig. 11. Sea clutter components removed by our proposed SWA block in Fig. 10, where (a) is the sea clutter image and (b) is the corresponding image spectrum.

Table V(a) indicates that the inclusion of 180 simulated wake images has a certain impact on the accuracy of detection. These real wake images enable Wake2Wake to concentrate on learning and suppressing wave features. We hypothesize that superior results could have been achieved with real wake images exhibiting more diverse sea conditions. Table V(b) outlines the impact of various convolution operations on wave suppression and detection results in our SWA block. Correspondingly, Fig. 10 provides a visual comparison of the results mentioned in Table V(b). The texture of the suppressed sea waves remains distinctly visible in the absence of either FcaConv or DSConv, as depicted in Fig. 10(b) and (g). When FcaConv is employed, the spectral characteristics in Fig. 10(h) indicate a slight increase in wave scale. This can be attributed to the absence of a significant Kelvin transverse wave, with only a divergent wave present on the left side. Consequently, FcaConv proves to have a limited contribution to this particular test sample. However, when both FcaConv and DSConv are employed, as depicted in Fig. 10(j), it is evident that the texture of the waves is effectively alleviated. This indicates that the simultaneous application of these two convolutions yields beneficial effects, aligning with the observation that both types of ship wakes are present in this figure simultaneously.

TABLE V

EFFECT OF DIFFERENT DATASETS USED TO TRAIN WAKE2WAKE AND VARIOUS CONVOLUTION OPERATIONS APPLIED IN SWA BLOCK ON DETECTION PERFORMANCE. THE RESULTS IN THIS TABLE REPRESENT DETECTION **RECALL** AND **MAP**, RESPECTIVELY. (a) COMPARISON OF THE EFFECT OF DIFFERENT SOURCES OF DATASET TO TRAIN WAKE2WAKE. HERE, OPENSARWAKE* REPRESENTS THE 65 IMAGES SELECTED FROM THE FULL OPENSARWAKE DATASET. (b) COMPARISON OF THE EFFECT OF DIFFERENT CONVOLUTION OPERATION UTILIZED IN OUR PROPOSED SWA BLOCK TO TRAIN WAKE2WAKE

Dataset		OBB Detector		
OpenSARWake*	Simulated Wake	Rotated RepPoints	S^2A-Net	Oriented R-CNN
✓		57.4/39.9	46.4/34.0	48.7/35.6
✓	✓	59.1/41.4	48.9/36.4	50.2/37.9

(a)

Dataset		OBB Detector		
FcaConv	DSConv	Rotated RepPoints	S^2A-Net	Oriented R-CNN
		54.9/36.8	44.0/31.9	45.4/34.2
✓		56.3/37.9	46.1/33.4	46.9/35.3
	✓	57.8/39.7	48.3/34.0	48.7/36.6
✓	✓	59.1/41.4	48.9/36.4	50.2/37.9

(b)

Furthermore, to visually illustrate the wave components removed in Fig. 10(e), we present in Fig. 11 the wave images suppressed by our proposed SWA block and the corresponding image spectrum. It can be observed that although a small portion of the wakes is suppressed along with the waves, the majority of the removed components are sea clutter.

IV. CONCLUSION

This study introduces Wake2Wake, a feature-guided self-supervised method for suppressing waves in SAR ship wake detection. Our proposed Wake2Wake utilizes a novel self-supervised training strategy. Instead of training directly on the full-scale dataset, we selected 65 real SAR images with the most prominent wave textures from the dataset and combined them with 180 simulated wake images. The simulated SAR images generated by our previous work exhibit a balanced

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:07:08 UTC from IEEE Xplore. Restrictions apply.

and ergodicity selection of radar parameters, sea conditions, and ship parameters, effectively compensating for the feature singularity of real SAR images in some scenarios. The results indicate that this training strategy produces increased detection accuracy compared to the original method. In addition, Wake2Wake employs a novel SWA block designed to capture the characteristics of turbulent and Kelvin wakes in SAR images. Results from experiments have proven the effectiveness of this block. Nevertheless, additional studies are required to achieve a tradeoff between wave suppression and wake detection. The next step involves merging the Wake2Wake framework with the OBB detector to create a real end-to-end network, following the approach illustrated in Fig. 1(b).

ACKNOWLEDGMENT

The authors would like to thank the editors who processed this article and the anonymous reviewers for their constructive comments toward improving it. They want to express our gratitude to the Alaska Satellite Facility and European Space Agency for the Sentinel, ALOS PALSAR, and TerraSAR-X data, available at https://search.asf.alaska.edu/#/ and https://earth.esa.int/eogateway/missions/terrasar-x-and-tandem-x.

REFERENCES

[1] X. Li et al., "Tropical cyclone morphology from spaceborne synthetic aperture radar," *Bull. Amer. Meteorol. Soc.*, vol. 94, no. 2, pp. 215–230, Feb. 2013.

[2] Y. Tang, S. Wang, J. Wei, Y. Zhao, and J. Lin, "Salient feature pyramid network for ship detection in SAR images," *IEEE Sensors J.*, vol. 24, no. 3, pp. 3036–3045, Dec. 2023.

[3] X. Zhang, S. Feng, C. Zhao, Z. Sun, S. Zhang, and K. Ji, "MGSFA-Net: Multiscale global scattering feature association network for SAR ship target recognition," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 4611–4625, 2024, doi: 10.1109/JSTARS.2024.3357171.

[4] X. Leng, K. Ji, S. Zhou, and X. Xing, "Ship detection based on complex signal kurtosis in single-channel SAR imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6447–6461, Sep. 2019.

[5] X. Leng, K. Ji, B. Xiong, and G. Kuang, "Complex signal kurtosis—Indicator of ship target signature in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5208312.

[6] Z. Sun et al., "An anchor-free detection method for ship targets in high-resolution SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 7799–7816, 2021.

[7] Z. Sun, X. Leng, Y. Lei, B. Xiong, K. Ji, and G. Kuang, "BiFA-YOLO: A novel YOLO-based method for arbitrary-oriented ship detection in high-resolution SAR images," *Remote Sens.*, vol. 13, no. 21, p. 4209, Oct. 2021.

[8] K. Ding et al., "Towards real-time detection of ships and wakes with lightweight deep learning model in Gaofen-3 SAR images," *Remote Sens. Environ.*, vol. 284, Jan. 2023, Art. no. 113345.

[9] Y. Guan, H. Xu, and C. Li, "A method of ship wake detection in SAR images based on reconstruction features and anomaly detector," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2023, pp. 6398–6401.

[10] J. Xin, J. Gu, J. Yang, Y. Wen, and D. Ding, "Ship wake detection in SAR images based on lightweight network," in *Proc. Int. Appl. Comput. Electromagn. Soc. Symp. (ACES-China)*, Aug. 2023, pp. 1–3.

[11] H. Xing, Y. Wang, J. Han, and J. Bai, "Research on ship wake detection method based on composite mode," in *Proc. 15th Int. Conf. Comput. Res. Develop. (ICCRD)*, Jan. 2023, pp. 290–297.

[12] Y. Yuan, Y. Wu, P. Feng, Y. Fu, and Y. Wu, "Segmentation-guided semantic-aware self-supervised denoising for SAR image," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5218416.

[13] K. Kang and D. Kim, "Ship velocity estimation from ship wakes detected using convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 11, pp. 4379–4388, Nov. 2019.

[14] K. Wang, X. Wang, L. Guo, J. Chen, and X. Yuan, "Radial velocity estimation approach of ship for spaceborne multi-channel HRWS SAR SLC data," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 59, no. 6, pp. 8710–8724, 2023, doi: 10.1109/TAES.2023.3309928.

[15] M. D. Graziano, G. Rufino, and M. D'Errico, "Wake-based ship route estimation in high-resolution SAR images," in *SAR Image Analysis, Modeling, and Techniques XIV*, vol. 9243. Bellingham, WA, USA: SPIE, 2014, pp. 220–231.

[16] O. M. Griffin, H. T. Wang, and G. A. Meadows, "Ship hull characteristics from surface wake synthetic aperture radar (SAR) imagery," *Ocean Eng.*, vol. 23, no. 5, pp. 363–383, Jul. 1996.

[17] F. Biondi, "A polarimetric extension of low-rank plus sparse decomposition and radon transform for ship wake detection in synthetic aperture radar images," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 1, pp. 75–79, Jan. 2019.

[18] A. Jiaqiu et al., "A novel ship wake CFAR detection algorithm based on SCR enhancement and normalized Hough transform," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 4, pp. 681–685, Jul. 2011.

[19] O. Karakuş, I. Rizaev, and A. Achim, "Ship wake detection in SAR images via sparse regularization," 2019, *arXiv:1904.03309*.

[20] F. Biondi, "Low-rank plus sparse decomposition and localized radon transform for ship-wake detection in synthetic aperture radar images," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 1, pp. 117–121, Jan. 2018.

[21] G. Yang, J. Yu, C. Xiao, and W. Sun, "Ship wake detection for SAR images with complex backgrounds based on morphological dictionary learning," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Mar. 2016, pp. 1896–1900.

[22] R. Del Prete, M. D. Graziano, and A. Renga, "First results on wake detection in SAR images by deep learning," *Remote Sens.*, vol. 13, no. 22, p. 4573, Nov. 2021.

[23] C. Li et al., "Detection-friendly dehazing: Object detection in real-world hazy scenes," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 7, pp. 8284–8295, 2023, doi: 10.1109/TPAMI.2023.3234976.

[24] S. Parrilli, M. Poderico, C. V. Angelino, and L. Verdoliva, "A nonlocal SAR image denoising algorithm based on LLMMSE wavelet shrinkage," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 2, pp. 606–616, Feb. 2011.

[25] F. Argenti and L. Alparone, "Speckle removal from SAR images in the undecimated wavelet domain," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 11, pp. 2363–2374, Nov. 2002.

[26] B. Xu, Y. Cui, Z. Li, B. Zuo, J. Yang, and J. Song, "Patch ordering-based SAR image despeckling via transform-domain filtering," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 4, pp. 1682–1695, Apr. 2014.

[27] T. Huang, S. Li, X. Jia, H. Lu, and J. Liu, "Neighbor2Neighbor: Self-supervised denoising from single noisy images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 14776–14785.

[28] A. Krull, T.-O. Buchholz, and F. Jug, "Noise2 void–learning denoising from single noisy images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2124–2132.

[29] Z. Wang, J. Liu, G. Li, and H. Han, "Blind2Unblind: Self-supervised image denoising with visible blind spots," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 2017–2026.

[30] X. Wang, W. Wang, Y. Cao, C. Shen, and T. Huang, "Images speak in images: A generalist painter for in-context visual learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 6830–6839.

[31] H. Kim, G. Papamakarios, and A. Mnih, "The Lipschitz constant of self-attention," in *Proc. 38th Int. Conf. Mach. Learn.*, 2021, pp. 5562–5571.

[32] F. Xue, W. Jin, S. Qiu, and J. Yang, "Rethinking automatic ship wake detection: State-of-the-art CNN-based wake detection via optical images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5613622.

[33] Z. Qin, P. Zhang, F. Wu, and X. Li, "FcaNet: Frequency channel attention networks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 763–772.

[34] C. Xu and X. Wang, "OpenSARWake: A large-scale SAR dataset for ship wake recognition with a feature refinement oriented detector," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[35] J. Staal, M. D. Abramoff, M. Niemeijer, M. A. Viergever, and B. van Ginneken, "Ridge-based vessel segmentation in color images of the retina," *IEEE Trans. Med. Imag.*, vol. 23, no. 4, pp. 501–509, Apr. 2004.

[36] V. Mnih, *Machine Learning for Aerial Image Labeling*. Toronto, ON, Canada: Univ. Toronto, 2013.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:07:08 UTC from IEEE Xplore. Restrictions apply.

[37] Y. Qi, Y. He, X. Qi, Y. Zhang, and G. Yang, "Dynamic snake convolution based on topological geometric constraints for tubular structure segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 6047–6056.

[38] M. Fazlyab, A. Robey, H. Hassani, M. Morari, and G. Pappas, "Efficient and accurate estimation of Lipschitz constants for deep neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 1–12.

[39] Q. Hua, Z. Yun, H. Mu, Y. Jiang, and D. Xu, "An approach of sea clutter suppression for SAR images by self-supervised complex-valued deep learning," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[40] S. Liu, Y. Lei, L. Zhang, B. Li, W. Hu, and Y.-D. Zhang, "MRD-DANet: A multiscale residual dense dual attention network for SAR image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5214213.

[41] C. Xu, R. Qi, X. Wang, and M. Tao, "Instability of energy spectrum disturbance for ship turbulent wakes: SAR imaging simulation and analysis," *Ocean Eng.*, vol. 292, Jan. 2024, Art. no. 116502.

[42] A. Wang, X. Wang, J. Chen, L. Guo, and H. Huang, "Wave spectrum retrieval method based on full-link ocean surface SAR imaging simulation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5219120.

[43] D. Martin, C. Fowlkes, D. Tal, and J. Malik, "A database of human segmented natural images and its application to evaluating segmentation algorithms and measuring ecological statistics," in *Proc. 8th IEEE Int. Conf. Comput. Vis.*, vol. 2, Jul. 2001, pp. 416–423.

[44] R. Franzen, "Kodak lossless true color image suite," vol. 4, no. 2, p. 9, 1999. [Online]. Available: http://r0k.us/graphics/kodak

[45] J. K. E. Tunaley, E. H. Buller, K. H. Wu, and M. T. Rey, "The simulation of the SAR image of a ship wake," *IEEE Trans. Geosci. Remote Sens.*, vol. 29, no. 1, pp. 149–156, Jan. 1991.

[46] G. Zilman, A. Zapolski, and M. Marom, "On detectability of a Ship's Kelvin wake in simulated SAR images of rough sea surface," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 2, pp. 609–619, Feb. 2015.

[47] I. G. Rizaev, O. Karakus, S. J. Hogan, and A. Achim, "Modeling and SAR imaging of the sea surface: A review of the state-of-the-art with simulations," *ISPRS J. Photogramm. Remote Sens.*, vol. 187, pp. 120–140, May 2022.

[48] D. Cozzolino, S. Parrilli, G. Scarpa, G. Poggi, and L. Verdoliva, "Fast adaptive nonlocal SAR despeckling," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 2, pp. 524–528, Feb. 2014.

[49] C.-A. Deledalle, L. Denis, and F. Tupin, "Iterative weighted maximum likelihood denoising with probabilistic patch-based weights," *IEEE Trans. Image Process.*, vol. 18, no. 12, pp. 2661–2672, Dec. 2009.

[50] G. Chierchia, D. Cozzolino, G. Poggi, and L. Verdoliva, "SAR image despeckling through convolutional neural networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 5438–5441.

[51] W. Li, Y. Chen, K. Hu, and J. Zhu, "Oriented RepPoints for aerial object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1819–1828.

[52] J. Han, J. Ding, J. Li, and G.-S. Xia, "Align deep features for oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5602511.

[53] X. Xie, G. Cheng, J. Wang, X. Yao, and J. Han, "Oriented R-CNN for object detection," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2021, pp. 3500–3509.

[54] H. Hersbach et al., "ERA5 hourly data on single levels from 1979 to present," *Copernicus Climate Change Service (C3S) Climate Data Store (CDS)*, vol. 10, no. 10, p. 24381, 2018.

[55] X. Ma, C. Wang, Z. Yin, and P. Wu, "SAR image despeckling by noisy reference-based deep learning method," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8807–8818, Dec. 2020.

[56] H. Feng, B. Hou, and M. Gong, "SAR image despeckling based on local homogeneous-region segmentation by using pixel-relativity measurement," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 7, pp. 2724–2737, Jul. 2011.



Chengji Xu (Member, IEEE) received the M.S. degree in software engineering from Shanghai Maritime University, Shanghai, China, in 2019. He is currently pursuing the Ph.D. degree in information and communication engineering with Sun Yat-sen University, Guangzhou, China.

His research interests include SAR ocean remote sensing, ship wake simulation, and ship wake detection.



Qingsong Wang received the Ph.D. degree from the National University of Defense Technology, Changsha, China, in 2011.

Since 2019, he has been with the School of Electronics and Communication Engineering, Sun Yat-sen University, Shenzhen, China, where he is mainly engaged in basic theory and engineering application research in the field of precision processing of microwave mapping. His research interests include multisystem radar positioning, image matching, and UAV navigation.



Xiaoqing Wang was born in Jiangxi, China, in 1978. He received the B.S. degree in electronic engineering from Xiamen University, Xiamen, China, in 2000, and the Ph.D. degree in signal and information processing from Chinese Academy of Sciences, Beijing, China, in 2005.

From 2005 to 2016, he was an Assistant Researcher and Associate Researcher at the Institute of Electronics, Chinese Academy of Sciences. From 2016 to 2019, he was an Associate Researcher and Researcher at the Institute of Microelectronics, Chinese Academy of Sciences. Since 2019, he has been a Professor with the School of Electronics and Communication, Sun Yat-sen University, Shenzhen, China. His research interests include ocean microwave remote sensing, SAR signal processing, and radar processing.



Xiaopeng Chao is currently pursuing the Ph.D. degree with the School of Electronics and Communication, Sun Yat-sen University, Shenzhen, China.

His research interests include ocean microwave remote sensing, synthetic aperture radar signal processing, deep learning, computer vision, and generative models.



Bo Pan was born in Hubei, China, in 1995. He received the B.S. degree in electrical engineering and automation from Hubei University of Technology, Wuhan, China, in 2016, and the M.S. degree in power electronics and power drives from the University of Electronic Science and Technology of China, Chengdu, China, in 2020. He is currently pursuing the Ph.D. degree in information and communication engineering with Sun Yat-sen University, Shenzhen, China.

His research interests include interferometric synthetic aperture radar (InSAR) and ocean remote sensing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:07:08 UTC from IEEE Xplore. Restrictions apply.